DOC-ANALOG-005

1 Summary and Physical Selection Criteria

In the architecture of an analog control computer, the operational amplifier that constitutes the error amplifier of a Type II compensator cannot be treated as an ideal abstract block with infinite gain. Nonlinear and parasitic imperfections in the input integrated circuit impose physical limitations that directly degrade the phase margin, introduce DC errors, and cause integration drifts over time.

The compensator's inverting summing node is a very high-impedance node connected to the R-C feedback network. To prevent the dynamics of the semiconductor itself from destabilizing the loop synthesized in the latest DOC issue, four strict numerical selection criteria are established:

- *Accuracy of the SPICE Macromodel:* Availability of an official, manufacturer-verified subcircuit for LTspice that accurately models the two-pole frequency response of the open-loop gain, the parasitic input capacitances, and the nonlinear current behavior.
- *Gain-Bandwidth Product:* At the target crossover frequency, the open-loop gain must exceed the gain required by the compensator by more than 50 dB, ensuring that the dominant pole of the amplifier introduces a phase delay of less than -0.2°.
- *Input Bias Current:* A bias current greater than 10 pA continuously injected into the integrator capacitor C_1 generates a residual voltage ramp that shifts the DC operating point outside the PWM modulation range.
- *Response Speed:* In response to step-load transients (10% to 90%), the control output must vary smoothly without suffering from rise-time distortion.

2 Comparative Evaluation Matrix for Operational Amplifiers

Four candidate components representing different manufacturing technologies (industrial-grade JFET, ultra-low leakage current CMOS, and standard laboratory JFET) are evaluated:

Critical Parameter	Loop Requirement	ADA4610-1 (ADI - JFET)	OPA140 (TI - JFET)	LMC6081 (TI - CMOS)	TL071 (Legacy JFET)
Availability of the Official SPICE Model	Essential	Yes (Integrated Native ADI)	Yes (LTI/TI SPICE)	Yes (PSpice Macro)	Yes (Generic)
GBW Product	≥ 10.0 MHz	16.3 MHz	11.0 MHz	1.3 MHz	3.0 MHz
Leakage Current (I_b)	≤ 10.0 pA	5.0 pA (máx)	10.0 pA (máx)	0.01 pA	65.0 pA
Slew Rate (SR)	≥5.0 V/µs	61.0 V/µs	20.0 V/µs	1.5 V/µs	13.0 V/µs
Offset voltage (V_{OS})	≤ 500 µV	400 µV (B Grade)	120 µV	350 µV	3000 µV
Thermal Drift ($\Delta V_{OS}/\Delta T$)	≤ 1.5 µV/ºC	1.00 µV/ºC	0.35 µV/ºC	1.00 µV/ºC	18.00 µV/ºC
DC Gain (A_{OL0})	≥ 100 dB	120 dB	136 dB	130 dB	100 dB

3 Justification for Selecting the ADA4610-1

The ADA4610-1 (a precision monolithic operational amplifier with a JFET input from Analog Devices) is selected as the definitive standard component for the analog computer for the following engineering reasons:

- *Native Integration in LTspice:* As a flagship device from Analog Devices, its macromodular-level subcircuit is fully integrated and optimized within the native LTspice library. The model includes secondary-pole transfer functions, common-mode and differential-mode capacitances, and the finite open-loop output impedance R_{out}, ensuring that pre-layout simulations accurately reflect the actual physics of the PCB.
- *Superior Phase Dynamics (*GBW *= 16.3 MHz):* Offers a bandwidth that is 48% greater than that of the OPA140, providing maximum gain margin at the crossover frequency of 10.0 kHz.
- *Response to Extreme Transients (*SR *= 61.0 V/µs):* Its high slew rate ensures that the loop responds to sudden load changes without entering a state of nonlinear saturation due to slew rate.

4 Analytical Verification of Imperfections and Dynamic Limits

4.1 Phase Delay Injected at the Crossover Frequency

The actual transfer function $A_{OL}(s)$ of the Type II compensator, incorporating the dominant-pole model of the op-amp, is expressed as:

$$A_{v.real}(s) = \frac{A_{v.ideal}(s)}{1 + \frac{1+A_{v.ideal}(s)}{A_{OL}(s)}}$$

At the target crossover frequency f_C = 10.0 kHz:

1. Available gain in open-loop mode for the ADA4610-1:

$$|A_{OL}(j2\pi{\cdot}10kHz)| = \frac{GBW}{f_C} = \frac{16.3 \times 10^6 Hz}{10.0 \times 10^3 Hz} = 1630 \ (64.24 \ dB)$$

2. Gain required by the Type II compensator (DOC-ANALOG-004):

$$|A_{v.ideal}(j2\pi{\cdot}10kHz)| = 2.3281 \ (7.34 \ dB)$$

3. Calculation of Phase Degradation (ϕ_{lag}):

$$\Delta\phi_{lag} \approx -arctan\left(\frac{|1 + A_{v.ideal}(f_C)|}{|A_{OL}(f_C)|}\right) = -arctan\left(\frac{1 + 2.3281}{1630}\right) = -arctan(0.0020418) \approx -0.117^{o}$$

4.2 Static Integration Derivation Using the Polarization Current

With the ADA4610-1's JFET input, the maximum guaranteed bias current is I_b = 5.0 pA. At the inverting summing node, this current flows into the zero-crossing capacitor (C_1 = 1.80 nF).

The static voltage ramp due to residual charge accumulation is calculated as follows:

$$\frac{dV}{dt} = \frac{I_b}{C_1} = \frac{5.0 \times 10^{-12} A}{1.80 \times 10^{-9} F} = 2.777 \times 10^{-3}\ V/s = 2.777\ mV/s$$

Since the analog loop operates with a response time of microseconds (100 µs), a drift of 2.78 µV per millisecond is completely negligible and imperceptible to the control system.

4.3 Evaluation of the Slew Rate

In the event of a transient disturbance that requires the PWM modulator to vary the control voltage V_{ctrl} within a range of ΔV_{ctrl} = 2.0 V during half a switching cycle (Δt = 5.0 µs):

$$Required\ rate\ of\ change = \frac{2.0\ V}{5.0\ \mu s} = 0.40\ \frac{V}{\mu s}$$

The ADA4610-1 has a slew rate of 61.0 V/µs, which represents a margin of 152.5 times the circuit's maximum demand.

5 Implications of PCB Layout and Parasitic Interaction

To maintain the integrity of the ADA4610-1 and prevent parasitic capacitances and impedances on the printed circuit board from affecting the zeros of the Type II network, the following physical design guidelines must be followed:

5.1 Guard Rings at the High-Impedance Input

Since the input resistance at the inverting pin of the ADA4610-1 is 10^{12} Ω, surface leakage currents on the PCB caused by flux residues or moisture can be on the order of nanoamperes (1,000 times greater than I_b).

- *Layout Rule:* A continuous analog ground trace must be routed to completely surround the inverting and non-inverting input pins of the ADA4610-1 on both layers of the PCB, forming a solderless guard ring.

5.2 Compensation for Input Parasitic Capacitance

The ADA4610-1 has a differential input capacitance $C_{in.diff}$ of approximately 2.8 pF and a common-mode capacitance $C_{in.cm}$ of approximately 3.8 pF.

- *Impacto en el Polo de Alta Frecuencia (f_{p1}):* The total parallel parasitic capacitance at the inverter node is $C_{para} \approx$ 6.6 pF. Since $C_{para} << C_2$ = 300 pF (2.2% of C_2), the position of the high-frequency pole f_{p1} = 26.42 kHz remains stable within the tolerances of discrete components.

5.3 Isolation of Bipolar Power Supplies

The ADA4610-1 will be powered by dedicated symmetrical ±12.0 V power rails, locally regulated by ultra-low-noise LDOs and decoupled at each power pin with a 100 nF C0G/NP0 ceramic capacitor in parallel with a 10 µF tantalum capacitor.

6 References and Sources of Contrast

- *Analog Devices. (2019).* ADA4610-1/ADA4610-2/ADA4610-4 Data Sheet: Low-Noise, Precision, Rail-to-Rail Output, JFET Dual/Quad Op-Amps. Analog Devices Technical Documentation.

- *Analog Devices. (2020).* LTspice Model Library: ADA4610-1 Macromodel Subcircuit Definition (v2.1). Analog Devices Software Distribution.

- *Jung, W. (2005).* Op Amp Applications Handbook. Analog Devices / Newnes.
  (Chapter 6: Passive Components and PCB Layout Techniques—Guard Rings and Parasitic Leakage).

- *Franco, S. (2014).* Design with Operational Amplifiers and Analog Integrated Circuits (4th ed.). McGraw-Hill Education.
  (Chapter 5: Dynamic Configurations and Finite Open-Loop Gain Bandwidth Effects).